GRENOBLE INP UGA Génie industriel

Data collection campaign: building a smell cartography

Smart Analytics for Big Data

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In this document, I will propose a data collection strategy of different types of data that can contribute to build a cartography of Grenoble, Saint Martin d'Hères and Echirolles according to the smells of its urban areas, quarter and streets.

The first type of data is the location of shops, restaurants, gardens and markets. It identifies the areas where pleasant smells are likely to occur, such as food odours, natural scents and fresh product smells. This can make a base layer of understanding the positive odours in the city.

The second type of data is the smell reports by the citizen. We can collect them with, for example, the SignalAir site. It provides a subjective information about how people perceive odours in their daily environment. It can indicate areas with persistent smells (pleasant / unpleasant) and can also show temporal patterns. This gives the human perception of the city odours with their impact on inhabitants. It can complement the objective measurements.

The third type of data is industrial emission reports. It identifies factories, treatment plants and other industrial activities that releases pollutants. Those pollutants could be associated with strong odours. It's important because Grenoble and its surroundings are highly industrialized, olfactory nuisance due to industrial activity is very likely to occur.

The fourth type of data is the air measurements that gives quantitative information about gases and particles in the atmosphere. These measurements give an objective confirmation of the presence and intensity of odours reported by citizens and it can also detect substances that may not be consciously notice but still influence air quality and comfort. They allow a scientific validation of smell hotspots and help to identify correlations between pollutants and reported perceptions. It is more interesting than the industry emission report as it can be almost continuously collected.

Finally, the data on waste locations and collection schedules contribute to understanding temporal variations in odours as garbage accumulation and collection are frequent sources of temporary but strong odours. These data help to explain why certain streets or areas may smell strongly at specific times or days. It can prevent misinterpretation of occasional events as permanent conditions.

Putting together all these data, allows us to create a cartography of smells of the city in several layers. The map could show zones with different smells, separating areas with good smells and those with bad smells. Hence, we can understand better how activities in the city make the smell identity of the streets, quarters or the city in general.

The Table 1 associates each type of data to the Big Data's Vs, its contribution to build the smell identity and its collection strategy.

Table 1. Data collection campaign table

Type of data	Associated V's	Contribution to build the smell ID (street, area, quarter, city)	Collection strategies
Location of shops, restaurants, gardens, market…	Volume: medium Velocity: slow (rarely updated) Variety: many activities Veracity: high (official) Value: low	Explanation of the odour's sources.	Collect data about Grenoble Metropole. Google maps API (costly).
Smell reports	Volume: variable Velocity: near real-time Variety: intensity, pleasant/unpleasant Veracity: subjective Value: low (cheap deployment)	Odour nuisance mapping.	Anyone can report odour nuisances on the website SignalAir. Collect this data.
Industrial emissions (reports)	Volume: medium Velocity: low (annual) Variety: pollutants, facilities Veracity: Value: high	Explanation of the odour's sources.	Collect environmental reports.
Air measurements	Volume: high (continuous sensors) Velocity: near real-time Variety: multiple pollutants Value: high (sensors, maintenance, calibration)	Direct identification of hotspots (streets) and temporality (quarter/city).	Use ATMO measurements data.
Waste location and collection schedule	Volume: low Velocity: low (weekly) Variety: services Veracity: high (official) Value: medium	Explains temporary odour events (garbage day).	Use public services calendar

ATMO site:

https://www.atmo-auvergnerhonealpes.fr/air-commune/Sassenage/38474/previsions?adresse=43%20Rue%20Francois%20Gerin%2038360%20Sassenage&date=2025-10-10

SignalAir site:

https://www.signalair.eu/fr/

Waste collection calendar of grenoble:

https://www.grenoblealpesmetropole.fr/12-jours-de-collecte-des-poubelles.htm